

Dylan Pierre

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PROFESSIONAL SUMMARY

Computer Science graduate with a strong foundation in backend systems, machine learning, and software security. Eager to apply a solid grounding in data structures, algorithms, and software engineering principles to a collaborative team environment. Quick learner who adapts readily to new tools, codebases, and engineering practices.

EDUCATION

Sonoma State University

Bachelor of Science in Computer Science

3.5 GPA

2025

- **Relevant Coursework:** Data Structures & Algorithms, Operating Systems, Database Systems, Computer Networks, Programming Languages, Software Engineering, Machine Learning

TECHNICAL PROJECTS

Authentication and Authorization Service | *Go, PostgreSQL, RESTful APIs, Backend Services, Security* | [\[GitHub\]](#)

- Designed and built a backend service in Go exposing RESTful API endpoints for authentication and authorization, consumed by external applications.
- Integrated PostgreSQL to persist user credentials and profile data, using pgx and normalized schema design.
- Implemented PASETO tokens and Argon2 password hashing to protect user credentials and prevent unauthorized access.
- Developed role-based access control middleware that authorizes every request, scoping data and actions to each user's permission level.

Cervical Spine Fracture Detection | *Python, Computer Vision, PyTorch* | [\[Link\]](#)

- Engineered a computer vision pipeline for fracture detection in medical CT imaging, preprocessing DICOM volumes with Hounsfield Unit (HU) windowing and building a balanced training set from the RSNA 2022 CT dataset.
- Trained a YOLOv8 model for per-vertebra localization via PyTorch Ultralytics for fracture identification, applying deep learning to DICOM and NIFTI CT data.
- Collaborated on a cross-functional team under a fast-paced timeline, running weekly stand-ups and delegating pipeline components (data preprocessing, detection, classification) to deliver a working end-to-end system.

C-Like Language Interpreter | *C++, Systems* | [\[GitHub\]](#)

- Designed and built a full language interpreter in C++ from scratch, taking source code from raw text to running output.
- Implemented every stage of the language pipeline: a state-driven tokenizer for lexical analysis, a recursive descent parser, an abstract syntax tree to represent program structure, a symbol table for scope resolution, and the tree-walking evaluator.

Redis-mini | *Go, Concurrency, Networking, Backend, Databases* | [\[GitHub\]](#)

- Built a concurrent, in-memory key-value database that accepts client connections over TCP, using goroutine-based request handling and read/write mutexes to support multiple simultaneous client sessions under concurrent load.
- Designed unit tests covering invalid input handling, state integrity, and concurrent access under load.

EXPERIENCE

DataAnnotation

Code Quality Analyst

May 2024 – Present

Remote

- Reviewed and debugged code across Python, JavaScript, C++, and Java, identifying logic bugs, performance issues, and security flaws.
- Evaluated API usage, edge cases, and error handling to ensure correctness against detailed technical specifications.

CERTIFICATIONS

CompTIA Security+ (Earned April 2026), **Google Cybersecurity Professional**

SKILLS

Programming Languages: C++, C#, Go, Python, Java, JavaScript, TypeScript, Rust, PowerShell, Bash

Tools & Methodologies: Git/Github, Agile, RESTful APIs, Object-Oriented Programming, CI/CD, VSCode

Databases: SQL, PostgreSQL, Redis

Cloud & Infrastructure: AWS, Docker, Virtual Machines

Operating Systems: Windows, Linux, macOS